

Sai Kiran

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SUMMARY

Software Engineer designing, building, and deploying scalable, cloud-native applications using Java, Python, and TypeScript. Proven track record shipping production-grade microservices, distributed data pipelines, and agentic AI systems at Morgan Stanley. Published ML researcher (EMNLP 2025) with deep expertise in LLMs, RAG, and multi-agent frameworks. Strong foundation in object-oriented design, data structures, algorithm design, and full SDLC practices including code reviews, test-driven development, and CI/CD.

WORK EXPERIENCE

Morgan Stanley | Software Engineer - Gen AI | Jul 2025 - Present

- Developed and maintained production Java/Spring Boot microservices processing 1M+ monthly transactions, debugged latency spikes and resolved production defects, reducing p95 response time to sub-200ms.
- Built and optimized Java 8 ETL pipelines processing 10M+ monthly records, resolved critical SQL query bottlenecks via indexing and query tuning, improving retrieval time by 35%.
- Built agentic AI workflows using Amazon Bedrock Agents and MCP tooling, deploying MCP servers on AWS Lambda and ECS with endpoints exposed via API Gateway; consolidated Jira, Confluence, and Bitbucket into a single agent interface, eliminating context-switching overhead across engineering workflows.
- Deployed and managed containerized microservices on Kubernetes (EKS), implemented CloudWatch-based observability and alerting for zero-downtime deployments, and tracked and resolved infrastructure issues across EKS clusters.
- Maintained code quality through Git-based source control, JUnit/PyTest unit and integration testing, peer code reviews, and thorough debugging of production issues; supported UAT and Preprod transitions, ensuring smooth delivery to enterprise clients.

University of Massachusetts Lowell | Research Assistant | Aug 2024 - Jun 2025

- Designed and developed a multi-agent LLM-based patient education platform using LLaMA 3.2-3B, LoRA fine-tuning, and PPO-based reinforcement learning trained on 10,000+ synthetic EHR clinical notes, achieving 5x BLEU improvement and 5% BERTScore gain over the GPT-4o-mini baseline.
- Engineered an automated data synthesis pipeline using GPT-4o-mini to generate 10,000 clinically diverse discharge notes and QA pairs across 14 disease categories, eliminating manual annotation overhead and reducing annotation cycle time by 30%.
- Authored technical documentation, QA reports, and experimental analyses, collaborated with global research teams on conversational healthcare AI, leading to an EMNLP 2025 publication, and delivered technical workshops and instruction, improving programming proficiency for 88 students and increasing engagement by 20%.

Hexaware Technologies | Software Engineer | Jul 2020 - Nov 2023

- Delivered a full-stack network management platform using React.js, TypeScript, and Node.js, architecting real-time monitoring dashboards for Wi-Fi infrastructure across airport and public venue deployments; improved network uptime by 35% and reduced operations team ticket volume by 22%.
- Implemented real-time data streaming via WebSocket connections, rendering live network performance and user traffic analytics through Chart.js and D3.js visualizations; reduced mean incident detection time by 20% and integrated REST APIs to feed live data into operational dashboards.
- Deployed containerized backend microservices on Docker and AWS ECS with auto-scaling policies, sustaining 99.9% API availability across a 6-month production window with zero unplanned downtime.
- Collaborated within Agile/Scrum teams across cross-functional squads, contributing to sprint planning, UAT cycles, and on-schedule production deployments while documenting system architecture, API contracts, and runbooks.

EDUCATION

Master of Science in Computer Science | University of Massachusetts Lowell

Jan 2024 - Dec 2025 | GPA: 3.9/4.0

Published "Chatbot to Help Patients Understand Their Health," EMNLP 2025 Findings.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, JavaScript/TypeScript, SQL, Bash/Shell

Backend: Spring Boot, Node.js, REST APIs, GraphQL, Microservices, gRPC

Databases: PostgreSQL, Redis, MongoDB, SQL Optimization, Indexing, Query Tuning

LLMs & Vector DBs: LLaMA, GPT-4, BERT, ChromaDB, PGVector, FAISS, Pinecone

Cloud & AWS: Lambda, ECS, EKS, S3, CloudWatch, API Gateway, SageMaker

Data Engineering: Spark, Hadoop, Kafka, Snowflake, PostgreSQL, Redis, Airflow

DevOps: Docker, Kubernetes, Jenkins, GitHub Actions, Terraform, CI/CD, Git

Methodologies: Agile/Scrum, Production Support, Observability, SOLID, SSDLC

OS & Tooling: Linux/Unix, Bash Scripting, Log Analysis

APIs & Standards: RESTful APIs, GraphQL, gRPC, OAuth2/JWT, Swagger/OpenAPI, API Gateway, MCP

PROJECTS

NowServing | Java 21, Spring Boot, React, TypeScript, PostgreSQL, Redis, WebSockets

- Developed a real-time virtual queue and reservation platform with customer queue tracking, QR-based joining, and an owner dashboard using Java 21, Spring Boot, React, TypeScript, PostgreSQL, and Redis.
- Implemented WebSocket updates, Redis Pub/Sub, database locking, and idempotent requests to maintain queue consistency; integrated TomTom travel estimates with venue service pace to calculate departure recommendations, with caching and fallback handling.

FitLive | React, TypeScript, Java 21, Spring Boot, PostgreSQL, SwiftUI

- Built a React/TypeScript fitness-planning application combining readiness check-ins, strength-session logging, deterministic progression, nutrition tracking, and pantry-aware shopping lists in a unified daily workflow.
- Implemented a Java 21/Spring Boot and PostgreSQL account backend with signed web requests, versioned updates, and duplicate-operation handling; developed SwiftUI and HealthKit integration with native device verification pending.

PUBLICATION

Chatbot To Help Patients Understand Their Health | EMNLP 2025 Findings (Author)

Designed and developed a multi-agent LLM system using LLaMA 3.2-3B, LoRA, and PPO-based RL, trained on 10,000 synthetic clinical notes, and improved BLEU by 5x and BERTScore by 5% over the GPT-4o-mini baseline, enabling patient-facing EHR comprehension at grade 7 readability, suitable for mobile deployment.